

MAT9004 Assignment 3

Due: Wednesday, 28 May 2025, 23:55 PM

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **one single pdf file** as other types might not display properly on moodle.

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1. [15 points] Consider the function $f(x, y) = (6x - x^2)(4y - y^2)$.

- (a) [1 point] Find the gradient $\nabla f(x, y)$.
- (b) [2 points] Find the Hessian matrix $H(x, y)$ for function $f(x, y)$.
- (c) [4 points] Locate and classify all stationary points of $f(x, y)$.
- (d) [1 point] Sketch the region $D = \{(x, y) : x + y \geq 4, x \leq 4, y \leq 5\}$.
- (e) [7 points] Find the global extrema of $f(x, y)$ over the domain D .

2. [18 points] **Scheduling appointments**

Four clients requested appointments at a service station in a given week. Each appointment is on one of the five working days (some appointments may be arranged during the same day). Answer the following questions.

- (a) [1 point] How many ways can one assign each of the four appointments to a day of the week?
 - (b) [4 points] How many arrangements from (a) are such that no day has more than one appointment? How many from (a) are such that no day has more than two appointments?
 - (c) [5 points] How many arrangements from (a) are such that there is an appointment on Monday? There are appointments on both Monday and Tuesday?
 - (d) [4 points] How many arrangements from (a) have exactly three days taken for appointments?
 - (e) [1 point] Let $(x_1, x_2, x_3, x_4, x_5)$ denote the workload profile, where x_i is the number of appointments on day i . How many different workload profiles are there for four appointments?
 - (f) [3 points] How many workload profiles from (e) are such that no day has more than two appointments?
3. [8 points] A fair coin is flipped six times. Assume the outcomes of the flips are independent. The outcomes of the coin flips form a palindrome if the sequence of T's and H's reads the same forwards and backwards, e.g. THTTHT. Let A denote the event that the first, second and fourth flips are all 'T'. Let Z denote the event that the six flips form a palindrome.
- (a) [4 points] Is A independent of Z ?
 - (b) [1 point] Let A^c be the complement of event A . Is A^c independent of Z ?
 - (c) [3 points] A fair coin is flipped six times and a certain property, Q , is being studied. Let B be the event that the first three flips are all 'heads'. It is found that $\mathbb{P}(Q|B) = 1/4$ and $\mathbb{P}(Q|B^c) = 2/7$. Find $\mathbb{P}(B|Q)$.
4. [6 points] Let X be a discrete random variable with probability distribution :

$$p_X(x) = \begin{cases} Kx^2 & \text{if } x = -3, -2, -1, 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

where K is a constant. Let $Y = |X|$.

- (a) [2 points] Find the value of K .
- (b) [4 points] Find the probability distribution of Y and hence deduce the expectation and variance of Y .

Total: 47 marks